iPerturb: A Biologically Informed and Explainable Framework for Context-Specific Single-Cell Perturbation Response Modelling

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Abstract. Predicting transcriptional responses to unseen genetic perturbations is a central task in single-cell biology, because perturbation effects depend on cellular context and spread through nonlinear regulatory interactions. Existing predictors rely on statistical associations or unconstrained graph learning, so they rarely encode regulatory mechanisms or explain predictions in biological terms. We propose iPerturb, a biologically informed and explainable framework that couples context-specific gene regulatory network (GRN) construction with learnable nonlinear perturbation propagation. First, iPerturb assembles a GRN per cell line from five evidence tiers, separating a signed, directed regulatory core from undirected association evidence and deriving response-based quantities from training perturbations only. Second, it propagates a clamped perturbation with regulator-shared Hill transfer functions, which have a provably unique steady state and run as matrix products, and integrates multi-hop propagation across tiers. Evaluated on 1,442 held-out CRISPRi target genes across two human cell lines, iPerturb builds cell-line-specific regulatory networks and outperforms a linear baseline and deep neural network–based methods (GEARS, CPA and biolord) on centred and magnitude-free correlation, top-20 correlation and centroid accuracy. Its Hill operator matches exact Hill propagation with several-fold less memory, and its signed paths predict the direction of first-hop responses, keeping every prediction interpretable at the level of named neighbours and signed regulatory interactions.

Keywords: gene expression prediction · genetic perturbation · gene regulatory network · biologically informed neural network · Hill function · interpretable AI

1 Introduction

Genome-scale CRISPR screens with single-cell read-out now measure the transcriptional response to thousands of genetic perturbations in a single experiment [30], yet every screen covers only a fraction of the genes, combinations and cell types of interest. Predicting the response to a perturbation that was never measured would let such screens be prioritised and extended *in silico*, and it

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is the task addressed by graph-based predictors such as GEARS [31]. A useful predictor must therefore generalise to unseen target genes, be specific to the cellular context, and explain why a gene is predicted to change [7].

The first obstacle lies in the network a predictor propagates over. GEARS propagates over a Gene Ontology graph and a co-expression graph [31], which encode functional association rather than direct regulation, and Genetic Neural Networks follow transcription-factor (TF) regulation but order their layers after removing feedback edges [12]. Curated collections such as CollecTRI provide signed TF–target interactions [26], but they are generic rather than cell-line-specific, cover TFs unevenly, and mix interactions with very different levels of evidence. A GRN for perturbation modelling has to combine heterogeneous evidence, keep the cell line in view, and record which edges are regulatory and which are merely associative.

The second obstacle lies in how a perturbation is propagated. Linear propagation, such as random walks over a network, is cheap and convergent but cannot represent saturation or the composition of activating and repressing inputs. Graph neural networks over a regulatory graph, as in SPECTRA [8], are flexible but learn their message functions rather than imposing a regulatory transfer law, and logic-based ODE models with Hill-type transfer functions give each edge its own parameters [11], which scales poorly to genome-wide networks. Because simple linear baselines remain hard to beat on unseen perturbations [2], nonlinearity has to be introduced with few parameters, a guaranteed steady state and an evaluation that controls for the network itself.

In this paper, we introduce iPerturb, a biologically informed and explainable framework that addresses both obstacles in two linked phases (Figure 1). First, iPerturb assembles a context-specific GRN for each cell line from five evidence tiers, from curated signed regulation to responses of training perturbations. Second, it propagates an unseen perturbation through this GRN: the perturbed gene is clamped, regulatory edges pass their input through a learnable Hill transfer function shared per regulator, and association edges carry observed training responses over multiple hops. The GRN of the first phase fixes *where* a perturbation can travel; the propagation of the second phase fixes *how* it travels and which edges explain each predicted change.

Our contributions are twofold.

- **Context-specific GRN construction.** We build a five-tier GRN per cell line that separates signed, directed regulation from undirected association evidence, derives all response-based edges from training perturbations only, and combines tier-specific multi-hop propagation with weights fitted for each cell line. On 1,442 held-out CRISPRi targets in K562 and RPE1, this raises the centred correlation of the predicted response from 0.145 to 0.319 (K562) and from 0.170 to 0.333 (RPE1) over a linear baseline, and the GRN outperforms its degree-preserving rewired version with the same read-out.
- **Learnable Hill-based regulatory propagation.** We propagate perturbations with regulator-shared Hill transfer functions under a persistent clamp, prove that the clamped steady state exists and is unique, and evaluate the

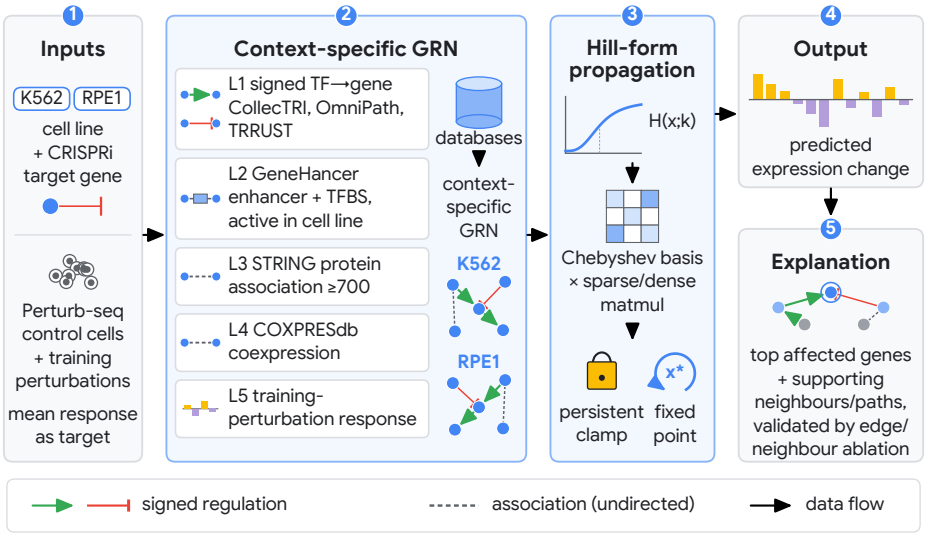


Fig. 1. Overview of the iPerturb framework. (1) A cell line and a CRISPRi target are the inputs; the response is the perturbation-conditioned mean expression change. (2) A context-specific GRN is assembled from five evidence tiers (L1 curated signed regulation; L2 enhancers and binding sites active in the cell line; L3 protein association; L4 co-expression; L5 training-perturbation response), with association evidence kept undirected and unsigned. (3) With the perturbed gene clamped, the perturbation is propagated over regulatory edges by a learnable Hill transfer function shared per regulator and evaluated as a matrix product, and observed training responses are propagated over association edges for multiple hops and integrated across tiers. (4) The predicted expression change is (5) explained by supporting neighbours and signed paths, each tested against controls. Graphs and bars are illustrative.

operator as a matrix product in a Chebyshev basis. It reproduces exact Hill propagation to 3.5×10^{-5}, uses 6.4- to 8.3-fold less memory on a dense graph, and its signed paths predict the direction of first-hop responses in 83% (K562) and 76% (RPE1) of responding genes.

Here, *explainable* is meant mechanistically: each predicted change decomposes into contributions of named neighbours and signed paths of the GRN, each supported by named databases; paths are candidate mechanisms, not validated molecular routes.

2 Related Work

Predicting responses to unseen perturbations. Latent-shift models represent a perturbation as a vector added in a learned latent space, as in scGen [25] and the compositional perturbation autoencoder CPA [24]; biolord disentangles

known cell attributes from the remaining variation [29]. These models generalise to an unseen gene only through a description of that gene, such as a gene-ontology embedding. GEARS learns gene and perturbation embeddings over a co-expression graph and a Gene Ontology graph and was designed for unseen single and combinatorial perturbations [31]. Foundation and language models pretrained on large cell atlases, scGPT [10], Cell2Sentence [19] and STATE [1], transfer learned cell-state representations to perturbation prediction, and SPECTRA predicts responses with a directed heterophilic graph neural network over an inferred GRN [8]. iPerturb differs from all of them in fixing the form of regulatory propagation (a Hill transfer under a clamp) and in separating regulatory from associative edges.

Evaluating perturbation-response models. Benchmarks show that deep perturbation models often fail to surpass a linear model on a low-rank gene embedding or the mean training response [2], and that a response shared by most perturbations inflates correlation-based metrics [37]. We therefore centre all responses on the training-mean response, report a magnitude-free correlation, compare against the linear low-rank baseline, and test every network against degree-preserving rewired and data-derived graphs with the same read-out.

Regulatory networks and their context. Curated resources collect signed TF–target interactions from the literature, including TRRUST [16], OmniPath [36] and CollecTRI [26]; GeneHancer links enhancers to target genes with tissue annotations [14]; STRING [33] and COXPRESdb [27] provide protein association and co-expression. Context-specific GRNs are usually inferred from single-cell multi-omics, as in SCENIC+ [6], and CellOracle uses such inferred GRNs to simulate TF knockouts by repeated matrix multiplication [17]. iPerturb instead combines public evidence tiers with the training perturbations of the same cell line, keeps association evidence undirected and unsigned, and simulates the perturbation with a persistent clamp whose steady state is proven to exist and to be unique.

Mechanistic and biologically informed models. Hill-type and thermodynamic models describe how the occupancy of a promoter by activators and repressors sets transcription [5,4]. Continuous logic models turn such transfer functions into ordinary differential equations on signalling networks [38], and cell-line-specific versions with a normalised Hill transfer have been fitted with one set of parameters per edge [11]; PHOENIX fits neural ODEs of gene expression with physiological priors [3]. Biologically informed neural networks shape their architecture by prior knowledge [32]: Genetic Neural Networks follow TF regulation with one layer per gene [12], and knowledge-primed and pathway-informed networks predict cell states or clinical phenotypes [15,13,23]. iPerturb shares the Hill parameter of a regulator across its targets, keeps feedback, and predicts genome-wide expression changes rather than a phenotype.

Network propagation. Propagating a signal over a biological network, by random walks with restart, diffusion kernels or truncated walks, amplifies weak signals by

integrating all paths between genes, and the normalisation of the walk controls the bias towards hubs [9]. iPerturb uses propagation twice: signed, clamped propagation of the perturbation over regulatory edges, and multi-hop propagation of observed training responses over association edges, integrated across evidence tiers. Learnable univariate functions in a basis expansion, whose evaluation reduces to matrix multiplication, underlie learnable-activation networks [21,20]; we use a Chebyshev expansion to keep Hill transfer functions cheap, including edge-specific shapes.

3 Data and Methods

3.1 Datasets

Public gene-regulation resources were curated from four complementary source types: (i) transcription factor (TF)–target interactions from TRRUST [16], CollecTRI [26] and OmniPath [36] (120,312 unique directed pairs, 61,215 of them with a defensible sign); (ii) enhancer–gene and TF-binding annotations from GeneHancer v5.26 [14]; (iii) protein–protein association pairs from STRING [33] filtered at combined score ≥ 700; and (iv) co-expression pairs from COXPRESdb [27]. Perturbation data are the genome-scale CRISPRi Perturb-seq screens of Replogle et al. [30] in K562 (chronic myelogenous leukaemia) and RPE1 (retinal pigment epithelium) cells. Starting from the raw UMI counts, each cell was normalised to 10^4 counts and log1p-transformed; the response of a perturbation p is the *mean-level* difference $\Delta_p = \overline{x}_p - \overline{x}_0$ between the mean log-normalised expression of its cells and of the control cells of the same gem groups (batch-matched), computed on the stored gene axis (8,563 genes in K562, 8,749 in RPE1). Perturbations with fewer than 20 cells and targets absent from the gene axis were removed, leaving 1,835 (K562) and 1,928 (RPE1) targets. The data are single-cell, but the predicted endpoint is the perturbation-conditioned mean response; cell-resolved generation is outside the scope of this work.

3.2 Preprocessing

Perturb-seq data are processed in three representations rather than one matrix: control cells for the baseline state, raw UMI counts for count-level effects, and perturbation effects with uncertainty. Cell quality control is applied per gem group with 5-median-absolute-deviation rules on log UMI and log detected genes (both sides) and on the mitochondrial fraction (upper side only; no universal mitochondrial cutoff, because knockdowns can change metabolism); cells carrying more than one guide target are excluded, and control cells are matched to perturbed cells by gem group. We apply no batch correction, no imputation and no cell-cycle regression, since each can remove or create perturbation-specific signal. QC kept 99.6% of the 310,385 K562 cells and 99.8% of the 247,914 RPE1 cells (1,246 and 501 cells failed the detected-gene rule; controls: 10,670 cells in 48 gem groups for K562, 11,475 in 56 for RPE1). The effect of a perturbation is

Algorithm 1 iPerturb: context-specific GRN assembly and integrated propagation

Require: evidence tiers $\mathcal{E}_1, \ldots, \mathcal{E}_4$; cell line c; training and validation perturbations with responses $(\mathcal{P}_{\mathrm{tr}}, \Delta_{\mathrm{tr}})$, $(\mathcal{P}_{\mathrm{va}}, \Delta_{\mathrm{va}})$; kernel grid $\mathcal{K}$; held-out target t
Ensure: predicted response $\hat{\Delta}_t$; signed paths and supporting neighbours of t

$A_\ell \leftarrow$ edges of $\mathcal{E}_\ell$ on the gene axis of c, Level 2 active in c ▷ *Phase I*
resolve duplicates by the lowest tier; set conflicting signs to unsigned
$R \leftarrow$ signed, directed edges of Level 1 ▷ regulatory core
$\mathcal{A} \leftarrow \{A_\ell\}$ as undirected, unsigned graphs ▷ association tiers
clamp $x_t \leftarrow x_t^0(1-\zeta)$ and iterate Eq. (2) on R ▷ *Phase II*
report the signed paths from t and their supporting databases ▷ Eq. (4)
$\hat{\Delta}^{(0)} \leftarrow$ linear low-rank prediction fitted on $\mathcal{P}_{\mathrm{tr}}$
for each tier graph $A_\ell \in \mathcal{A}$ **do**
  for each kernel $\kappa \in \mathcal{K}$ and degree correction γ **do**
    $S \leftarrow \kappa(A_\ell)$; $\hat{\Delta}_p^{(\ell)} \leftarrow \sum_{j \in \mathcal{P}_{\mathrm{tr}} \setminus \{p\}} S_{pj} \deg_j^{-\gamma} \Delta_j \,/\, \sum_j S_{pj} \deg_j^{-\gamma}$
  keep the (κ, γ) with lowest validation error; $\hat{\Delta}_p^{(\ell)} \leftarrow \hat{\Delta}_p^{(0)}$ if p reaches no training target
$w \leftarrow \mathrm{NNLS}\big([\hat{\Delta}^{(0)}, \hat{\Delta}^{(1)}, \ldots, \hat{\Delta}^{(L)}]_{\mathcal{P}_{\mathrm{va}}}, \Delta_{\mathrm{va}}\big)$; $w \leftarrow w / \sum_\ell w_\ell$
return $\hat{\Delta}_t = \sum_\ell w_\ell \hat{\Delta}_t^{(\ell)}$ and the top-weighted neighbours of t

the difference of mean log-normalised expression to the batch-matched controls (Δ, Section 3.1) with a Welch standard error, a pseudobulk $\log_2$ fold change computed from summed UMIs (library-size normalised, pseudocount), and the knockdown efficiency of the target, $\hat{k}_p = 1 - \bar{x}_{p,\mathrm{target}} / \bar{x}_{\mathrm{control,target}}$ on the linear scale, reported only when the target is detected in controls (≥ 5 CPM); otherwise it is left unknown rather than set to complete knockdown. Cells and guides are not treated as independent replicates: the standard error is descriptive and is used for response labels and precision weights only. Gene sets, graph edges, loadings and all hyper-parameters are fitted on training perturbations only; the 8k-gene axis is the full stored gene axis of the screen (targets are kept even when they are not highly variable).

3.3 Constructing the context-specific GRN

To construct the GRN of a cell line, we integrated the evidence of Section 3.1 into a five-tier hierarchy ordered by evidence strength. Level 1 forms the signed, directed backbone: curated TF–target interactions from TRRUST, OmniPath and CollecTRI, extended with transcriptional edges from SIGNOR [22]; an edge keeps a sign only when its sources agree. Level 2 adds unsigned TF–target edges whose regulatory element is active in the cell line: GeneHancer enhancers with a tissue annotation of the line and a TF-binding site of the TF, and, as an extension, the 20 strongest factors per target gene in ChIP-Atlas experiments of the same line (K562 or hTERT RPE-1, ±5 kb of the transcription start site) [39]. Level 3 adds protein association, STRING pairs with combined score ≥ 700, CORUM complex co-membership [35] and BioGRID physical interactions [28],

and Level 4 adds co-expression: at most 20 of the 50 top-ranked COXPRESdb partners of each gene, and the 50 strongest partners per gene in GTEx samples re-aligned by ARCHS4 [34,18]. Level 5 is the observed response of the training perturbations of the line; it enters the GRN as the signal that is propagated (Section 3.4) and is recomputed for every fold, so no held-out response reaches the network. Levels 3–4 are association evidence: they are stored as undirected, unsigned edges and never drawn as regulatory arrows.

Cross-tier duplicates were resolved in favour of the lowest-numbered tier, and within-tier sign conflicts defaulted to unsigned, so an unknown sign is never promoted to activation. Context therefore enters at three points: the gene axis and Level 2 depend on the cell line, Level 5 comes from that line's training perturbations, and the weights that combine the tiers are fitted on that line's validation perturbations. All tiers are restricted to the gene axis of the screen, and each tier is kept as a separate graph A_ℓ so that its contribution can be measured and weighted. Algorithm 1 summarises how the tiers are assembled and propagated; the propagation operators are defined in Section 3.4.

3.4 Learnable propagation that follows the Hill function

Hill-form regulatory propagation. Let $x_i \in (\varepsilon, 1-\varepsilon)$ be the regulatory activity of gene i and $x_i^0 = \sigma(z_i)$ its baseline, where z_i is the standardised control expression (a fixed monotone map, so the graph cannot be bypassed by a learned encoder). A regulator j passes its activity through a modified Hill transfer function

$$H(x; k, n) = \frac{k^n \left[1 - (1-x)^n\right]}{(1-x)^n + k^n}, \qquad H(0) = 0,\ H(1) = 1, \tag{1}$$

(the TF4 form of the logic-ODE literature [38,11]) with $n = 2$ and one learnable $k_j \in [0.05, 1]$ *per regulator*, shared by all of its outgoing edges. Signed messages are baseline-centred, so an unperturbed cell produces exactly zero response:

$$x_i^{t+1} = (1-\eta)\, x_i^t + \eta\, \sigma\Big(\operatorname{logit} x_i^0 + \sum_{j \to i} w_{ji} \left[H(x_j^t; k_j) - H(x_j^0; k_j)\right]\Big), \tag{2}$$

with $w_{ji} = \rho\, s_{ji}\, c_{ji}\, \mathrm{softplus}(a) / \max\{1, \sum_{j'} |\cdot|\}$, where $s_{ji} \in \{-1, +1\}$ is the curated sign, c_{ji} the evidence confidence, and ρ, a are learnable scalars. A CRISPRi knockdown of gene p is a *persistent clamp*, $x_p^t = (1-\zeta) x_p^0$ at every iteration, where ζ is the median knockdown of the training perturbations. The fixed point x^* is found by damped iteration ($\eta = 0.5$, 24 steps) and the predicted response is $\hat{\Delta}_g = d_g\,(x_g^* - x_g^0)$ with a non-negative diagonal decoder d_g; no dense $G \times G$ decoder is used, so every prediction decomposes over named edges.

Matrix-multiplication form. Per-edge Hill evaluation requires gathering x_j for every edge and scattering messages, which maps poorly to accelerators. We therefore approximate each regulator's Hill curve by a Chebyshev expansion of degree r,

$$H(x; k_j) \approx \sum_{b=0}^{r} a_b(k_j)\, T_b(2x - 1), \tag{3}$$

whose coefficients $a_b(k)$ are least-squares fitted to Eq. (1) on a logarithmic grid of k and linearly interpolated in $\log k$, so that k_j stays learnable and differentiable. With $\mathbf{h} = \hat{H}(x) - \hat{H}(x^0) \in \mathbb{R}^{B\times G}$ for a batch of B perturbations, the aggregation in Eq. (2) is one sparse (or dense) matrix product $\mathbf{h}\,\mathbf{W}$, with $\mathbf{W}_{ji} = w_{ji}$, instead of an edge-wise gather and index-add. The same fixed-point equations are solved; only the arithmetic changes (equivalence is verified in Section 4.3).

Clamped steady state and multi-hop explanation. For explanations the same network is read as a signed linear system around the baseline. With the perturbed gene g clamped and all other genes R free,

$$\Delta x_R = (I - W_{RR})^{-1} W_{Rg}\, \Delta x_g = \sum_{k\geq 1} W_{RR}^{k-1} W_{Rg}\, \Delta x_g, \tag{4}$$

where W_{ij} is the signed, confidence-weighted effect of gene j on gene i (activation > 0, repression < 0, unknown signs excluded) and Δx_g is the log-expression change implied by the partial knockdown (training-median efficiency, not a complete knockout). The k-th term is the contribution of regulatory walks of length k, so every predicted change decomposes into hops and into signed paths; the strongest paths to a gene are reported together with the databases that support each edge and with the reinforcing and opposing path mass, so that cancellation is visible. We scale W so that $\rho(|W|) = 0.9$; Theorem 2 in Appendix E shows that this makes the multi-hop series converge absolutely for every clamped gene (and Theorem 1 gives a unique convergence point for the nonlinear Hill-form iteration, whose gain is parametrised so that it is a contraction for every parameter value). Paths are candidate mechanisms, not proven causal routes.

Response propagation over the association network. The association channel generalises the neighbour mean to multi-hop propagation of *training responses* (network propagation [9]). Seeds are training targets carrying their observed response vectors, and a test gene p receives $\hat{\Delta}_p = \sum_j S_{pj}\Delta_j / \sum_j S_{pj}$ over training targets j, where S is a propagation kernel of the association network: a random walk with restart, $S = \alpha(I - (1-\alpha)W)^{-1}$ with $W = AD^{-1}$ or $D^{-1/2}AD^{-1/2}$, a heat kernel e^{-tL}, or a walk truncated after K hops, with an optional degree-bias correction (division by $\deg_j^{\gamma}$) and a blend weight with the linear baseline selected on validation perturbations; the kernels converge for every $\alpha \in (0, 1]$ and give convex combinations of training responses (Proposition 3). Networks of different tiers can be propagated separately and combined with non-negative weights fitted on validation perturbations.

Association channel. Most perturbed genes are not curated TFs. Let $\mathbf{G} \in \mathbb{R}^{G\times k}$ be the top-k gene loadings of the centred training responses, and $\hat{A}$ the row-normalised adjacency of the (undirected) context-specific GRN. The embedding of a perturbed gene t is smoothed over its neighbours, $e_t = (1-\alpha)\mathbf{G}_t + \alpha\,(\hat{A}\mathbf{G})_t$, and the response is $\hat{\Delta}_t = \bar{\Delta} + e_t\,\mathbf{W}_e\,\mathbf{G}^\top$ with a ridge-fitted $\mathbf{W}_e$; k and α are chosen on the validation perturbations from fixed grids. For $\alpha = 0$ this is the standard linear low-rank baseline, so the channel nests it.

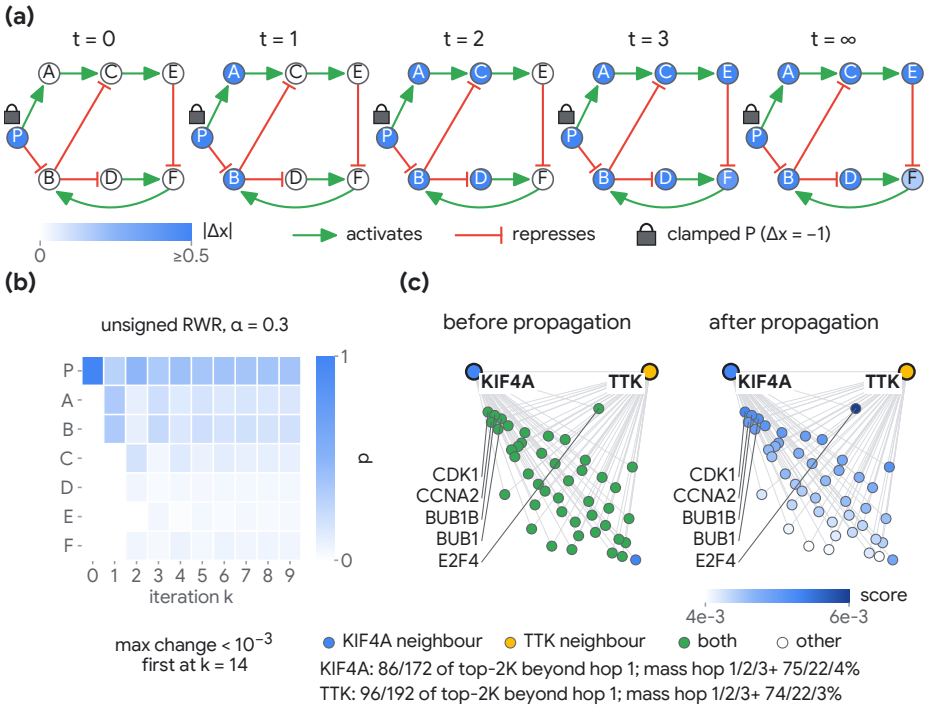


Fig. 2. Multi-hop propagation in iPerturb. **(a)** A CRISPRi perturbation (clamped gene) propagates over a small signed network; node colour is the predicted change after $t = 1, 2, 3$ hops and at the convergence point. **(b)** The association channel propagates signal over unsigned edges by a random walk with restart. **(c)** On the K562 association network, propagation from two genes assigns mass beyond their direct neighbours.

SCC-aware execution. Feedback makes Eq. (2) cyclic. For a frozen graph we compute strongly connected components once, order their condensation topologically, restrict the computation to the nodes reachable from the perturbed gene, settle acyclic singletons with a single forward update and iterate feedback components locally with their upstream inputs fixed. The predictions are those of the global solver (Section 4.3); this is an execution strategy, not a different model.

3.5 Evaluating iPerturb

For each cell line, targets were split into training, validation and test perturbations (70/15/15%) in three folds stratified only by properties known without response data (TF status, control-expression tertile, curated out-degree bin); no target is shared across the splits of a fold. All graph-derived quantities, loadings,

knockdown efficacy and hyper-parameters use training (and validation) perturbations only. Because the mean response of a perturbation is dominated by a response shared across perturbations [37], the primary metric is the Pearson correlation of Δ *centred* on the training-mean response (ρ_c), computed per perturbation without the perturbed gene and averaged over perturbations; we also report the same correlation after projecting the training-mean direction out of both profiles (ρ_r, which removes response magnitude), the top-20 centred correlation (ρ_{20}), the mean normalised rank of the correct perturbation among all test perturbations (Cent.; 0.5 is chance) and the area under the precision–recall curve for differentially expressed genes (|Welch z| > 4, cell level, no biological replicates, prevalence-dominated by the shared response). Intervals are 95% bootstrap intervals over unique held-out targets (each target averaged over the folds in which it is tested); paired differences use the same bootstrap. We compare against no change, the training-mean response, a ridge model on a low-rank gene embedding [2], GEARS [31], CPA [24] and biolord [29] trained on the same folds, released checkpoints of scGPT [10] and STATE [1] (Appendix C), a two-layer graph convolutional network on the same GRN, and control GRNs with degree-preserving rewired edges or a graph learned from the training data alone. The full protocol was fixed in a hashed pre-registration file before fitting, and later analyses are labelled as such in the repository.

4 Results

4.1 iPerturb builds cell-line-specific GRNs around a reliable signed core

To assess whether the GRN reflects the cell line, we compared the cell-line tier (Level 2) across lines. Restricted to enhancers active in each line and to TFs whose binding sites match, the GeneHancer tier contains 645,570 TF–target pairs for K562, 335,734 for HepG2 and 80,414 for Jurkat, and the edge sets overlap little (Jaccard index 0.08 K562–HepG2, 0.07 K562–Jurkat, 0.05 HepG2–Jurkat); TF out-degrees agree only moderately between lines (Spearman $\rho = 0.31$ to 0.71). For RPE1, whose cell type has no GeneHancer TF-binding annotation, the cell-line tier comes from ChIP-Atlas experiments in hTERT RPE-1 cells. The signed core is reliable: on training perturbations, the sign of a curated edge predicts the direction of the knockdown response of its target in 80% (K562; 109 responsive edges) and 68% (RPE1; 222) of cases (binomial $p < 4 \times 10^{-8}$; a per-TF test over the 8 and 9 TFs with at least five responsive edges gives $p =$ 0.008 and 0.012).

The preprocessing of Section 3.2 keeps the responses stable (median correlation with responses computed without per-gem-group QC 1.0000 in K562 and 1.0000 in RPE1) and records that knockdown is partial: the median efficiency over targets with a detectable knockdown is 84% (K562, 1,835 targets) and 91% (RPE1, 1,927), which motivates the partial clamp of Eq. (4).



Fig. 3. iPerturb predicts responses to unseen CRISPRi targets in two human cell lines. Mean over held-out targets with 95% bootstrap intervals over unique targets (three target-held-out folds; K562 $n = 699$, RPE1 $n = 743$). Filled circles, K562; open squares, RPE1. **(a)** Centred correlation ρ_c (primary; the training mean scores 0 by construction). **(b)** Correlation after projecting the training-mean direction out of both profiles (ρ_r). **(c)** Top-20 centred correlation. **(d)** Mean normalised rank of the correct perturbation (centroid accuracy; 0.5 is chance). Rewired-GRN and data-kNN graphs are controls with the same read-out and tuning grid.

4.2 iPerturb accurately predicts responses to unseen perturbations

Because the mean response of a knockdown is dominated by a component shared across perturbations, raw correlation is saturated by the training-mean response (0.26 in K562 and 0.49 in RPE1); all comparisons therefore use centred correlations (Figure 3, Table 1).

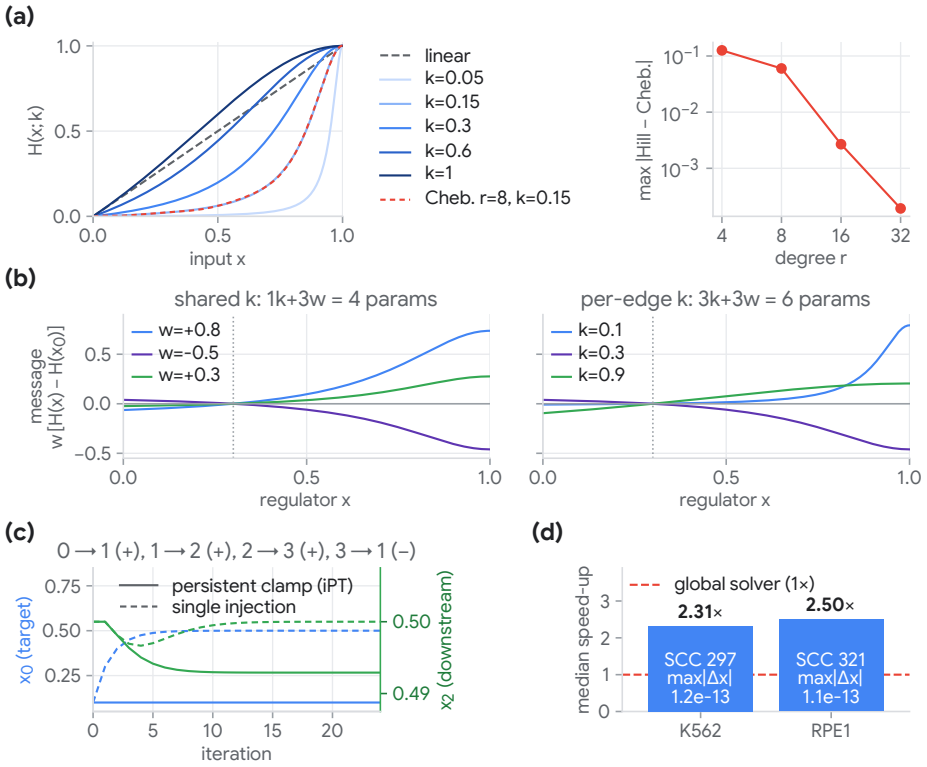
iPerturb, with multi-hop propagation integrated across evidence tiers, reaches $\rho_c = 0.319$ in K562 and 0.333 in RPE1, a gain of $+0.174$ [$+0.155$, $+0.194$] and $+0.163$ [$+0.133$, $+0.193$] over the linear low-rank baseline. The gain holds for every metric in Table 1, including the magnitude-free correlation ρ_r and the centroid rank, so it reflects the shape of the response rather than its size. iPerturb exceeds GEARS [31] trained on the same folds ($\rho_c = 0.042$ and 0.163) and

Table 1. iPerturb outperforms baselines and controls on held-out CRISPRi targets. Unique-target means over three folds (bold: best per column among held-out models). ρ_c: centred correlation; ρ_r: after removing the training-mean direction; ρ_{20}: top-20 centred correlation; Cent.: mean normalised centroid rank; AUPRC: differentially expressed genes (|Welch z| > 4). †Released pretrained checkpoint evaluated without retraining; its pretraining data overlap with these screens. CPA (B7) and biolord (B8) use the official packages with gene-ontology embeddings for unseen targets (Appendix C); external models are scored on the targets they support.

	K562 (699 targets)					**RPE1** (743 targets)				
Model	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC
Baselines										
No change (B0)	0.096	0.000	0.116	0.503	0.099	0.115	0.000	0.050	0.498	**0.245**
Train mean (B1)	0.000	0.000	0.000	0.000	0.031	0.000	0.000	0.000	0.000	0.054
Linear low-rank (B2)	0.145	0.139	0.275	0.641	0.110	0.170	0.108	0.276	0.634	0.175
External models										
GEARS (B4)	0.042	0.028	0.089	0.569	0.056	0.163	0.008	0.257	0.608	0.151
CPA (B7)	-0.001	-0.002	-0.004	0.488	0.041	0.002	0.000	0.012	0.514	0.066
biolord (B8)	0.001	0.005	0.004	0.497	0.059	0.000	0.002	-0.004	0.503	0.079
scGPT, zero-shot KO†	0.020	-0.004	-0.015	0.511	0.061	0.030	0.006	-0.053	0.510	0.097
STATE, zero-shot†	0.157	0.105	0.347	0.818	0.070	0.186	0.088	0.372	0.751	0.115
STATE, few-shot†	0.253	0.185	0.472	0.882	0.100	0.343	0.151	0.528	0.816	0.143
Controls (same read-out)										
Rewired GRN	0.141	0.134	0.268	0.633	0.110	0.163	0.108	0.260	0.631	0.174
Data-kNN graph	0.147	0.141	0.277	0.640	0.112	0.172	0.109	0.271	0.630	0.175
STRING only	0.235	0.223	0.430	0.735	0.132	0.250	0.184	0.375	0.679	0.192
iPerturb read-outs of the GRN										
Smoothing	0.238	0.221	0.434	0.737	0.132	0.252	0.179	0.374	0.680	0.191
GCN (B5)	0.267	0.240	0.454	0.755	0.140	0.299	0.201	0.422	0.711	0.191
Neighbour mean	0.285	0.276	0.511	0.791	0.162	0.302	0.254	0.440	0.737	0.204
Multi-hop (RWR)	0.291	0.286	0.520	0.805	0.169	0.316	0.267	0.461	0.752	0.205
Tier-integrated	**0.319**	**0.308**	**0.570**	**0.824**	**0.180**	**0.333**	**0.282**	**0.472**	**0.752**	0.210

CPA and biolord with gene-ontology descriptions of unseen targets, which stay close to the mean response. Released checkpoints of scGPT and STATE, whose pretraining overlaps with these screens, are listed for reference († in Table 1).

The gain comes from the GRN, not from the capacity of the read-out. With the same read-out and tuning grid, the GRN improves on its degree-preserving rewired version by +0.097 [+0.081, +0.113] (K562) and +0.089 [+0.066, +0.113] (RPE1) and on a graph learned from the training responses alone by +0.091 [+0.076, +0.107] and +0.080 [+0.057, +0.104]; run over rewired tier graphs, the tier-integrated propagation falls back to the linear baseline ($\rho_c = 0.143$ in K562; the real GRN is better by +0.176 [+0.157, +0.196]); zeroing each training row's own-target entry leaves the gain unchanged (+0.098 [+0.084, +0.113] and +0.089 [+0.066, +0.113]; Table 2). Every read-out of the GRN improves on the linear baseline, from one-step smoothing of gene embeddings (+0.093 [+0.077, +0.108] and +0.082 [+0.059, +0.106]) to a graph convolutional network and the neighbour mean, and reading the graph over multiple hops adds a further +0.008 [+0.002, +0.015] (K562) and +0.015 [+0.007, +0.024] (RPE1) over one hop (Section 5.1).



Speed-up (SCC-aware vs global solver) is measured in a numpy/python reference implementation, not an optimized kernel.

Fig. 4. The regulatory operator of iPerturb. (a) Modified Hill transfer functions of Eq. (1) for five dissociation constants, the linear transfer, and the degree-8 Chebyshev form for $k = 0.15$; the inset shows the maximal approximation error against the degree. **(b)** Messages of one regulator to three targets with a shared k (rescaled copies of one shape) or edge-specific k (three shapes). **(c)** A persistent clamp re-imposes the perturbed gene at every update, whereas a single injection lets it relax (four-gene motif with feedback; x_2 on the right axis). **(d)** Median speed-up of the SCC-aware solver over the global solver.

4.3 The learnable Hill propagation is faithful to the Hill form and efficient as a matrix product

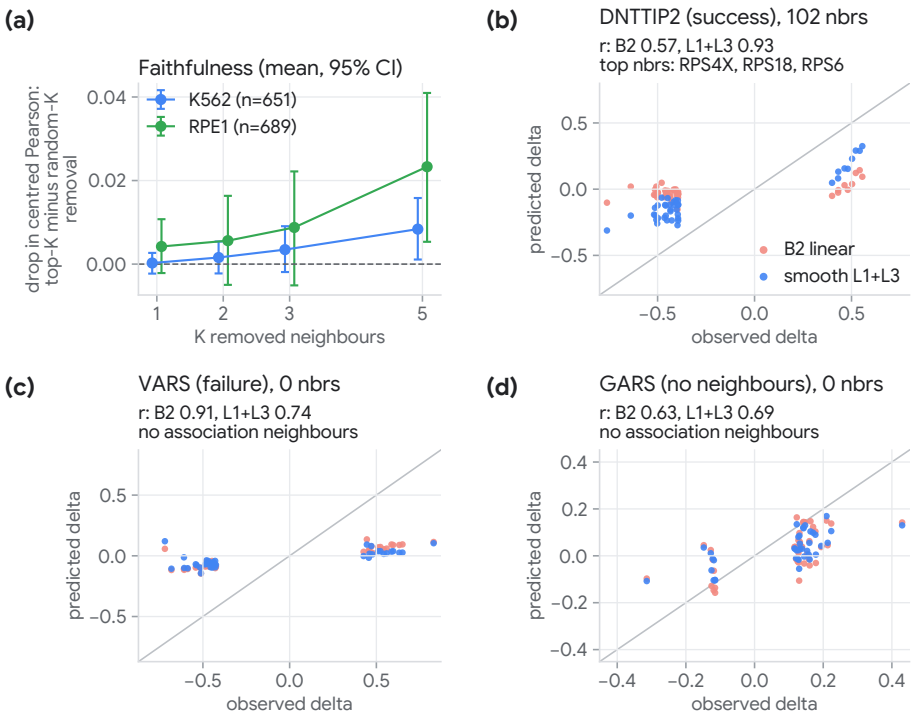
The Chebyshev form approximates the Hill curve to 2.7×10^{-3} at degree 16 (worst case over $k \in [0.05, 1]$; Figure 4a). Inside the full fixed-point iteration on the curated K562 network with heterogeneous regulator constants and 200 clamped targets, the equilibrium differs from exact Hill propagation by at most 3.5×10^{-5} at degree 16, and sparse and dense products agree exactly. The SCC-aware solver reproduces the global solver to 1×10^{-13} with a median speed-up of 2.3× (K562) and 2.5× (RPE1; Figure 4d).

Aggregating messages as a matrix product makes propagation efficient on dense evidence: on the 247k-edge association network (A100, batch 128, 24 damped updates) it reduces the step time from 250 ms to 77 ms and the peak memory from 6949 MB to 1078 MB, i.e. 3.3- to 4.6-fold faster with 6.4- to 8.3-fold less memory across batch sizes (Figure 8c). The Chebyshev form extends this to edge-specific learnable shapes: edge-wise evaluation needs activations of size $B \times E \times r$ and exceeds the 40 GB device at batch 128 with $r = 16$, whereas the matrix form applies one sparse matrix with Er non-zeros and fits in 9.9 GB (6.2-fold less memory at $r = 8$; outputs agree to 1×10^{-7}; Figure 8d).

4.4 iPerturb explains predictions by named neighbours and signed regulatory paths

Each prediction is attributed to the neighbours that support it, by the change in the prediction when a neighbour is removed, without access to the observed response. Removing the five most attributed neighbours lowers the centred correlation with the observed profile more than removing five random neighbours, by +0.008 [+0.001, +0.016] in K562 (651 targets) and +0.023 [+0.005, +0.041] in RPE1 (689; Figure 5a). The largest gain in the pre-specified K562 fold is DNTTIP2, a nucleolar protein whose three most attributed neighbours are the ribosomal proteins RPS4X, RPS18 and RPS6 (Figure 5b). Propagation recovers such functional modules more generally (Figure 6): all 30 top-scoring neighbours of the held-out ribosomal gene RPL11 are ribosomal proteins, and predicted profiles of test perturbations cluster by gene family (86% of edges between annotated genes join the same family, against 21% by chance).

On the regulatory side, we simulated CRISPRi as a persistent partial clamp on the signed network (Eq. (4)) and compared the predicted direction with the measured response of held-out targets that have regulatory out-edges (31 K562 and 47 RPE1 test perturbations; Figure 7). Among responding genes ($|z| > 3$) one hop downstream, the signed paths predict the direction correctly for 0.83 [0.75, 0.89] in K562 (114 genes) and 0.76 [0.71, 0.80] in RPE1 (358 genes), above the same network with permuted signs (0.73 and 0.70) or rewired edges (0.69 and 0.58). The controls lie above 0.5 because most responding genes move in the same direction after a knockdown, so the comparison isolates the information in the signs. In the reported case, SSRP1 in K562 acts through MYC on ribosomal-protein genes along activating edges supported by CollecTRI and OmniPath, and the predicted down-regulation matches three of the five most responsive genes shown (Figure 7c). Signed paths longer than one hop are reported as candidate mechanisms with their supporting databases (Section 5.4).



Case rule (K562, fold 0): gain = Pearson(L1+L3) - Pearson(B2); (b) max gain, (c) min gain,
(d) alphabetically first test target without neighbours.
Scatter: 40 genes with largest |observed| delta, perturbed gene excluded.

Fig. 5. Neighbour explanations are faithful. (a) Removing the top-K attributed neighbours of a test target lowers its agreement with the observed profile more than removing K random neighbours (mean difference with 95% interval; targets with at least six neighbours). **(b–d)** Cases chosen by a rule fixed beforehand (K562 fold 0): largest gain over the linear baseline, smallest gain, and the first target without neighbours. Neighbours are association evidence, not causal paths.

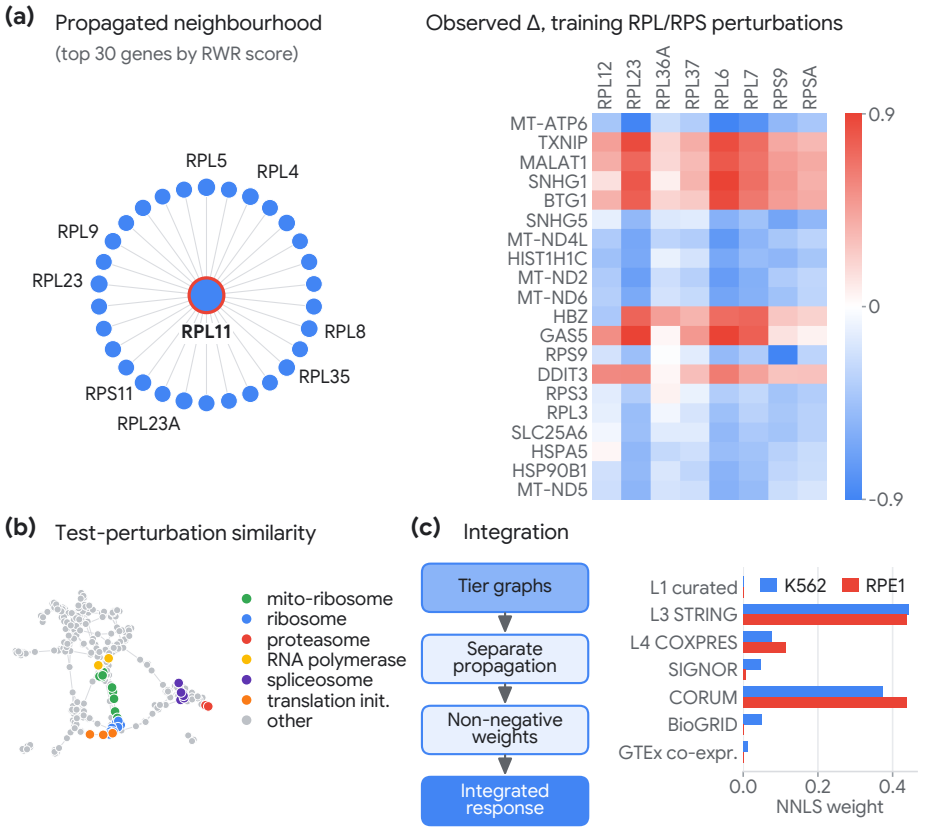


Fig. 6. Propagation over the association network recovers functional modules. (a) Random-walk-with-restart neighbourhood of the held-out ribosomal gene RPL11 (K562, fold 0; node area proportional to propagation score) and the observed responses of eight training ribosomal perturbations, whose profiles correlate at 0.52 on average. **(b)** Test perturbations joined to their three most similar predicted profiles; edges between family-annotated nodes join the same family in 86.0% of 43 cases against 21.1% expected by chance. **(c)** Integration of tier-specific propagations with non-negative weights (mean over three folds); the protein-association tiers carry most of the weight.

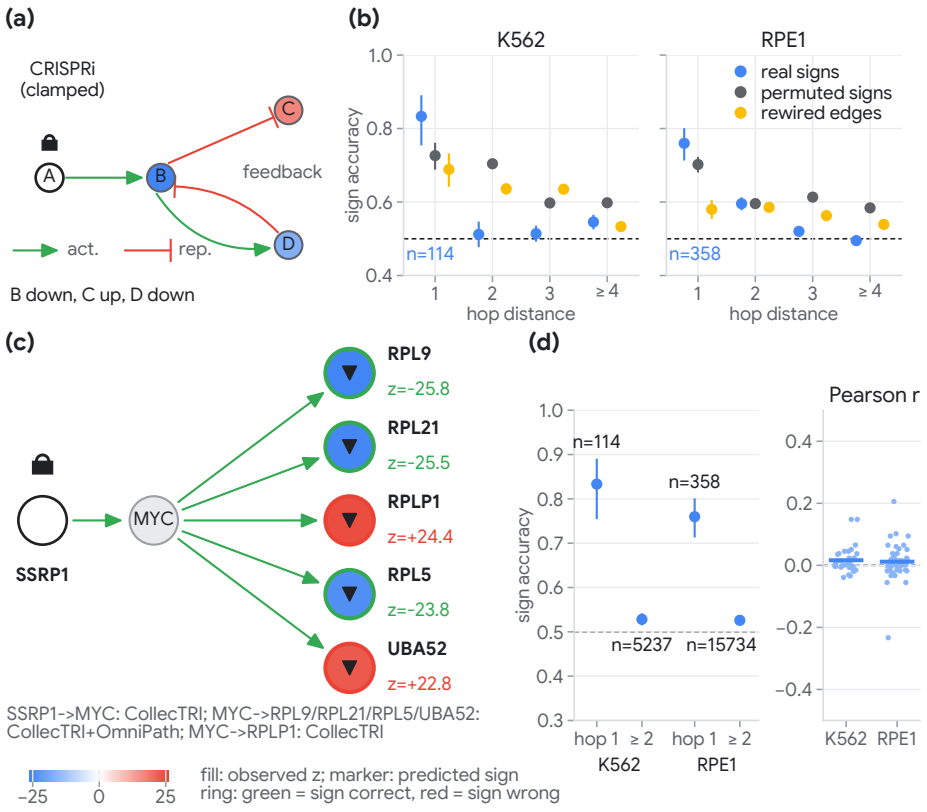


Fig. 7. Signed regulatory paths explain first-hop responses. (a) A CRISPRi perturbation of A clamps A and lowers B, raises C (repressed by B) and lowers D, as computed by the model. **(b)** Sign accuracy of the predicted direction among responding genes ($|z| > 3$) of held-out targets, by regulatory distance, for the real network and for controls with permuted edge signs or rewired edges (Wilson 95% intervals, pooled over draws). **(c)** A mechanism report for one held-out K562 target: predicted sign (marker) against observed response (fill), supporting databases below. **(d)** Real-network accuracy at the first hop versus later hops, and the correlation between predicted and observed responses across reached genes.

Table 2. Paired differences in ρ_c between iPerturb read-outs and controls. Mean difference with 95% bootstrap interval over unique held-out targets.

Comparison (ρ_c, mean difference [95% CI])	K562	RPE1
GRN vs no graph	+0.096 [+0.082, +0.110]	+0.087 [+0.064, +0.110]
GRN vs rewired GRN	+0.097 [+0.081, +0.113]	+0.089 [+0.066, +0.113]
GRN vs data-kNN graph	+0.091 [+0.076, +0.107]	+0.080 [+0.057, +0.104]
GRN vs rewired (self-entries zeroed)	+0.098 [+0.084, +0.113]	+0.089 [+0.066, +0.113]
iPerturb vs linear (B2)	+0.093 [+0.077, +0.108]	+0.082 [+0.059, +0.106]
iPerturb vs GEARS	+0.196 [+0.177, +0.215]	+0.090 [+0.059, +0.120]

5 Ablation Study

We ablate the two components of iPerturb, the context-specific GRN (evidence tiers, propagation depth, tier integration, cell-line tier) and the learnable Hill propagation (transfer function, certified variant, depth of signed paths), and the preprocessing. All differences are paired over unique held-out targets with 95% bootstrap intervals (Table 2).

5.1 Evidence tiers and multi-hop propagation

To measure what each tier contributes, we propagated the observed training responses over each tier alone (Table 3), in the manner of network propagation [9]. Protein-association evidence carries the predictive signal: STRING alone reaches $\rho_c = 0.298$ (K562) and 0.313 (RPE1), and CORUM complexes and BioGRID interactions are informative on their own, whereas the curated regulatory tier alone does not improve on the linear baseline (0.145 and 0.170) and the co-expression tiers improve on it only modestly; adding signed SIGNOR edges to the curated tier changes ρ_c by +0.015 [+0.006, +0.025] (K562) and +0.013 [-0.002, +0.030] (RPE1). Reading the curated-plus-STRING graph beyond the first hop improves on the one-hop neighbour mean, by +0.008 [+0.002, +0.015] (K562) and +0.015 [+0.007, +0.024] (RPE1) for random walks with restart and by +0.017 [+0.011, +0.023] and +0.020 [+0.008, +0.031] for walks truncated after two or three steps. Propagating each tier separately and integrating the read-outs with non-negative weights adds a further +0.029 [+0.021, +0.036] (K562) and +0.016 [+0.002, +0.030] (RPE1). The fitted weights concentrate on protein association (K562: STRING 0.44, CORUM 0.37, BioGRID 0.05, COXPRESdb 0.08; RPE1: STRING 0.44, CORUM 0.44; Figure 6c), so the curated tier contributes explanation rather than prediction. Figure 8b summarises the matched controls: only the real association evidence changes ρ_c relative to no graph, while rewired and data-derived graphs with the same read-out do not.

5.2 Cell-line-specific tier

We tested in K562 whether the cell-line tier carries signal specific to its own line, with a criterion registered before the run: the K562 GeneHancer tier must beat the HepG2 and Jurkat tiers when added to the curated-plus-STRING graph.

Table 3. Each evidence tier as a propagation graph, and multi-hop read-outs. Unique-target mean ρ_c over three target-held-out folds. The response of a held-out target is the propagated, normalised response of training targets over the graph; the restart probability or the number of hops is chosen on validation targets.

Graph (random walk with restart, α selected)	K562 ρ_c	RPE1 ρ_c
Linear low-rank (B2)	0.145	0.170
L1 curated (CollecTRI/OmniPath/TRRUST)	0.131	0.162
L1 + SIGNOR	0.146	0.175
L3 STRING	0.298	0.313
L3 CORUM complexes	0.263	0.278
L3 BioGRID physical	0.241	0.264
L4 COXPRESdb	0.173	0.208
L4 GTEx co-expression	0.170	0.195
L2 ChIP-Atlas (cell line)	0.130	–
L1+L3, one hop (neighbour mean)	0.282	0.301
L1+L3, random walk with restart (multi-hop)	0.291	0.316
L1+L3, truncated 2–3 hops	0.299	0.321
All tiers, propagated separately and integrated	0.319	0.333

Adding the K562 tier changes ρ_c by -0.011 [-0.019, -0.004] relative to the base graph, by -0.005 [-0.011, 0.000] relative to the HepG2 tier and by -0.008 [-0.015, 0.000] relative to the Jurkat tier, so the criterion was not met; the enlarged graph remains better than its rewired version by +0.064 [+0.052, +0.076]. The cell-line tier thus shapes the network and its explanations (Section 4.1), while the predictive signal is carried by association evidence that is shared between lines; the ChIP-Atlas tier behaves alike (Table 3).

5.3 Hill versus linear transfer

We trained the regulator-shared Hill model and a linear model with the same graph, decoder and budget as a residual on the linear baseline, on the targets reachable from the signed core (37 K562 and 50 RPE1 unique targets; Figure 8a). The two transfer functions predict equally well: the Hill form differs from the linear form by +0.0014 [-0.0001, +0.0041] (K562) and +0.0006 [0.0000, +0.0018] (RPE1), and from the no-graph baseline by +0.0001 [-0.0003, +0.0007] and -0.0004 [-0.0015, +0.0002]. The same holds for a certified variant with a log-odds state, a leaky Hill transfer and a gain that is a contraction for every parameter value (Appendix E), trained with a mean-squared or a correlation loss (+0.0001 [-0.0016, +0.0020] and -0.0024 [-0.0086, +0.0016] in K562, -0.0003 [-0.0014, +0.0006] and -0.0006 [-0.0026, +0.0010] in RPE1, relative to no graph; Table 5). An oracle read-out with its scale fitted on the test targets adds nothing, so the limit is the reach of the signed core (5.8–7.1% of perturbed genes have a signed out-edge) rather than the transfer function or the optimiser. The Hill operator therefore serves as the faithful, convergent and efficient regulatory model of iPerturb and as its explanation channel, while prediction is carried by the association tiers.

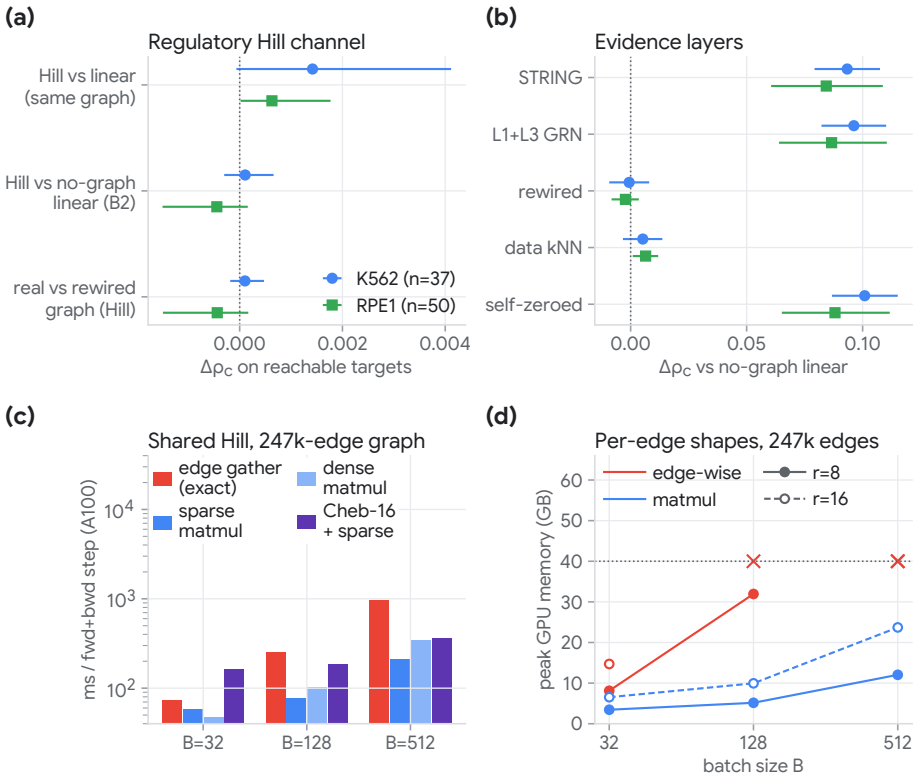


Fig. 8. Ablations and efficiency. (a) Hill-form versus linear propagation on the signed core, reachable test targets only (differences in ρ_c; 37 K562 and 50 RPE1 unique targets). **(b)** Change in ρ_c relative to the no-graph linear model for evidence layers and controls (STRING only; L1+L3; rewired; graph learned from training data; L1+L3 with own-target entries zeroed). **(c)** Time per forward and backward step of the regulator-shared Hill operator on the 247k-edge association network (A100). **(d)** Peak memory for edge-specific Chebyshev shapes (crosses: out of memory on 40 GB).

5.4 Depth of signed paths

The sign accuracy of the clamped steady state decreases with regulatory distance: at two hops it is 0.51 (K562) and 0.60 (RPE1), at three hops 0.51 and 0.52, and at four or more hops 0.55 and 0.49, at the level of the permuted-sign control (Figure 7b,d). Responses far from the perturbed gene are dominated by shared stress and growth programmes, which the association tiers capture; signed paths beyond the first hop are therefore reported as candidate mechanisms.

5.5 Preprocessing

Weighting training perturbations by cell number or by response precision changes ρ_c by -0.005 [-0.008, -0.003] and -0.010 [-0.014, -0.006] in K562 and +0.021 [-

0.002, +0.043] and +0.005 [-0.022, +0.033] in RPE1, so the headline models are unweighted.

6 Conclusion and Discussion

We introduced iPerturb, which builds a cell-line-specific regulatory network from five evidence tiers and propagates perturbations through it with a learnable transfer function that follows the Hill function, so that every prediction can be traced to named genes and edges. On held-out CRISPRi targets in K562 and RPE1 the tiered network carries predictive signal that matched rewired and data-derived graphs do not, multi-hop propagation of training responses over the tiers, integrated across tiers, is the strongest read-out (0.319 and 0.333 in ρ_c against 0.145 and 0.170 for the linear baseline), and the Hill-form operator reproduces exact Hill propagation to 3.5×10^{-5} while its matrix-product form is memory-lean on dense graphs. Two limitations are central. First, the tiers that carry the predictive gain (curated regulation and protein association) are the same for every cell line; the cell-line-specific enhancer tier, testable only in K562, differs strongly between lines but did not improve prediction over a foreign line's tier or over the base graph, so its context specificity is descriptive rather than predictive in this setting. Second, on the curated signed core, which reaches only 5.8–7.1% of the perturbed genes in these essential-gene screens, a nonlinear Hill transfer does not predict better than a linear one, and the gain over a linear low-rank model comes from association evidence rather than from curated regulation. The mechanism-level test sharpens this: the signed network predicts the direction of direct regulatory effects, but the signed multi-hop cascade is not supported beyond the first hop in these endpoint data, so we present longer paths only as candidate mechanisms, and we prove only that the multi-hop quantity is well defined and has a unique convergence point (Appendix E), not that it is correct. We read the coverage limitation as a statement about the screens, not about biology: the screens perturb mostly non-TF genes, and a signed core cannot propagate what it does not reach.

Several limitations qualify these results. The predicted endpoint is the mean response, not a distribution of cells. RNA abundance is a proxy for regulator activity and endpoint data cannot identify time constants, so the fixed-point operator is a steady-state model; neighbour attributions are tested by ablation but are association paths, not validated mechanisms. The folds and seeds reuse the same biological screens, and graph and no-graph models were tuned on comparable but not identical budgets; matched controls share their grid. The largest gains are obtained for genes in well-characterised complexes (for example, ribosomal-protein neighbours), which are enriched in essential-gene screens, and the transfer to screens with few such complexes is untested. The GeneHancer tier is empty for RPE1, so the cell-line-specific tier is tested in K562 only; a test with the training-perturbation tier (Level 5) is left to future work. Future work will add more cell lines and a genome-wide screen with more transcription-factor

targets, and test edge-specific learnable shapes, which the matrix-multiplication form now makes affordable, on data in which a signed core has more reach.

Conflict of interests, acknowledgments, funding, and data availability

Conflict of Interests. The authors declare that they have no competing interests.

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Data and Software Availability. The CRISPRi Perturb-seq datasets analysed in this study are publicly available from Replogle et al. [30]. Prior biological knowledge was obtained from TRRUST, CollecTRI, OmniPath, GeneHancer v5.26, STRING v12 and COXPRESdb; GeneHancer is licence-gated and is not redistributed. The source code, the frozen pre-registration of the evaluation protocol with its later addenda, figure-generation scripts and the files from which every number in this paper is generated accompany the submission and will be released in the public iPerturb repository (https://github.com/kagtgi/iPerturb) at publication.

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A Additional protocol details

Splits and fitting. Three target-held-out folds per line (70/15/15%, stratified by TF status, control-expression tertile and curated out-degree bin; seed 42, fold-specific seeds). The smoothing head selects $k \in \{5, 10, 20, 40\}$ and $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on validation perturbations (selected settings: K562, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75; RPE1, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75); the ridge penalty is 10^{-2}. The GCN has two layers, hidden size 64, residual self-weights, AdamW (learning rate 10^{-3} or 3×10^{-3}, weight decay 10^{-3}), at most 200 epochs, patience 20, three seeds. Hill-form models use $n = 2$, damping 0.5, 24 updates, learning rates $\{0.003, 0.01\}$ and penalties $\{10^{-3}, 10^{-1}\}$ on the per-gene decoder offset, selected on validation reachable targets; the clamp strength ζ is the median training knockdown.

Pre-registration. The evaluation protocol was written and hashed before any graph model was fitted. Later analyses (association smoothing, controls, GCN, explanations, context-specific tier, efficiency study, neighbour-free fallback) were added as dated addenda before they were run; the fallback analysis is exploratory and is not part of the headline numbers.

GEARS. Trained with the official package on at most 100 cells per perturbation and 3,000 control cells (log1p of counts per 10^4), for 6 epochs (fewer than the package default, which exceeded the memory of our 83 GB host at the full cell count), hidden size 64, with its default gene-ontology graph; targets absent from its ontology set are excluded from its evaluation. Its predictions are expressed relative to its own control mean, whereas the observed responses use batch-matched controls, so GEARS is centred on the mean of its predictions for a sample of 300 training targets before scoring (the raw correlation is reported in the repository). These settings make GEARS a budget-limited baseline.

B Efficiency measurements

On the CPU, the edge-wise form was the fastest for the sparse curated graph at every batch size (387 ms at batch 128 against 1485 ms for the sparse product), and on the association graph the dense product matched it (3477 ms vs. 3580 ms).

Table 4. Time per forward and backward step (ms) and peak memory (MB) of the regulator-shared Hill operator (24 updates, A100 40 GB). E0: exact Hill, edge gather/scatter; E1: exact Hill, sparse matrix product; E2: dense product; C1: degree-16 Chebyshev form, sparse product.

Graph	Batch	E0	E1	E2	C1
curated, 10k edges	128	37 / 942	72 / 1054	104 / 1592	179 / 4001
curated, 10k edges	512	73 / 3708	151 / 3315	346 / 3791	305 / 15018
association, 247k edges	128	250 / 6949	77 / 1078	103 / 1606	185 / 4025
association, 247k edges	512	960 / 27689	208 / 3338	346 / 3803	360 / 15041

C External baselines

All external models were scored with the folds, metrics and centring of the other models (predictions are centred on the mean prediction for training targets). *STATE*: the released State Transition checkpoints for Replogle-HVG data (zero-shot and few-shot variants, K562 and RPE1), evaluated without retraining; their pretraining data overlap with these screens (marked †). *scGPT*: the whole-human checkpoint without fine-tuning; the official perturbation head is trained from scratch, so we ran a zero-shot in silico knockout (target expression set to zero in 100 control cells, change of the predicted expression averaged over cells) on the 1,199 most expressed in-vocabulary genes; targets outside the vocabulary or with near-zero control expression are excluded. *biolord*: official package, with the gene-ontology neighbour vector of the GEARS gene-ontology graph as the attribute of the perturbed gene; no tuning. *CPA*: `cpa-tools` with the same neighbour vectors as frozen perturbation embeddings, the hyper-parameters of its Norman tutorial, at most 50 cells per training target and a wall-clock cap of 25 minutes per fold (a reduced budget).

D Data replay

An independent re-computation of the target of ten random K562 perturbations directly from the raw counts (no chunking or sparse accumulation) reproduced the stored responses to within 4.5×10^{-8} (single-precision rounding); the script and its output are in the repository.

Table 5. Certified, anti-vanishing Hill propagation on reachable held-out targets. Unique-target mean ρ_c and paired difference to the no-graph linear baseline (95% bootstrap interval); all variants use the same graph, decoder, optimiser and tuning grid. Predictions are not numerically collapsed (standard deviation of the learned correction 10^{-3} against a residual standard deviation of about 0.06–0.09), but they carry no usable signal.

Reachable targets	K562 ρ_c	Δ vs B2	RPE1 ρ_c	Δ vs B2
No graph (B2)	0.073	ref.	0.147	ref.
Linear propagation, MSE	0.069	-0.0041 [-0.0139, +0.0017]	0.146	-0.0012 [-0.0045, +0.0010]
Linear propagation, correlation loss	0.067	-0.0055 [-0.0161, +0.0018]	0.145	-0.0022 [-0.0051, -0.0001]
Certified Hill, MSE	0.073	+0.0001 [-0.0016, +0.0020]	0.147	-0.0003 [-0.0014, +0.0006]
Certified Hill, correlation loss	0.070	-0.0024 [-0.0086, +0.0016]	0.146	-0.0006 [-0.0026, +0.0010]
Certified Hill, correlation + Huber	0.070	-0.0024 [-0.0087, +0.0016]	0.146	-0.0006 [-0.0026, +0.0010]

E Convergence of multi-hop propagation

We show that the multi-hop propagation used by iPerturb has a unique convergence point (a fixed point) that is reached geometrically from any starting state, and that the hop-by-hop decomposition used for explanations is well defined. Throughout, $W \in \mathbb{R}^{G\times G}$ holds the signed effect of gene j on gene i (W_{ij}), g is the clamped (perturbed) gene, R the set of free genes, $\|W\|_\infty = \max_i \sum_j |W_{ij}|$ is the largest total incoming weight, $\rho(\cdot)$ the spectral radius and $|W|$ the entry-wise absolute value. Convergence is a property of the operator; it does not make a prediction correct (Remark 2).

Operators. (O1) *linear:* $\Delta x_R = W_{RR}\Delta x_R + W_{Rg}\Delta x_g$. (O2/O3) *Hill-form:* with a state z_R ($z = \operatorname{logit} x$; the sigmoid-state variant iterates x directly), baseline z^0 and clamped z_g,

$$F_i(z) = z_i^0 + \sum_j W_{ij}\big[T(\sigma(z_j)) - T(\sigma(z_j^0))\big], \qquad T(x) = (1-\lambda)H(x;k_j) + \lambda x, \quad (5)$$

iterated as $z^{t+1} = (1-\eta)z^t + \eta F(z^t)$ with the clamped coordinate held fixed at every step.

Lemma 1 (Hill slope). *For $n = 2$ and $k \le \sqrt{3}$, $L_H(k) := \max_{x\in[0,1]} |H'(x;k)| = \dfrac{9(1+k^2)}{8\sqrt{3}\,k}$, attained at $1 - x = k/\sqrt{3}$. With leak λ, $L_T \le (1-\lambda)L_H(k) + \lambda$.*

Proof. With $u = 1-x$, $H = k^2(1-u^2)/(u^2+k^2)$ and $|H'| = 2u\,k^2(1+k^2)/(u^2+k^2)^2$. The derivative of $u/(u^2+k^2)^2$ has the sign of $k^2 - 3u^2$, so the maximum is at $u = k/\sqrt{3} \le 1$, where $u^2 + k^2 = 4k^2/3$ and $|H'| = 9(1+k^2)/(8\sqrt{3}k)$. The leak adds slope λ and removes at most λL_H.

Numerically $L_H = 13.02$ at $k = 0.05$, 2.36 at $k = 0.3$ and 1.30 at $k = 1$; the formula agrees with a dense-grid maximum in the test suite.

Proposition 1 (Existence). *For every W, every parameter value and every clamp, the map (5) has a fixed point.*

Proof. T takes values in $[0,1]$, so each bracket in (5) lies in $[-1,1]$ and F maps $\mathbb{R}^R$ into the compact convex box $\{|z_i - z_i^0| \le \|W\|_\infty\}$; F is continuous, and Brouwer's theorem applies. (For the sigmoid state the box is $[0,1]^R$.)

Theorem 1 (Uniqueness and geometric convergence). *Let $q := \frac{L_T}{4}\|W_{RR}\|_\infty < 1$ and $q_\eta := 1 - \eta(1-q)$. Then the damped clamped iteration is a contraction of $(\mathbb{R}^R, \|\cdot\|_\infty)$ with constant q_η. It has a unique fixed point z^*, which is also the unique fixed point of F, and for every start z^0 and all t,*

$$\|z^t - z^*\|_\infty \le q_\eta^t \|z^0 - z^*\|_\infty, \qquad \|z^t - z^*\|_\infty \le \frac{q_\eta}{1-q_\eta}\|z^t - z^{t-1}\|_\infty.$$

A tolerance τ is reached after $t \ge \ln\big(\tau(1-q_\eta)/\|z^1 - z^0\|_\infty\big)/\ln q_\eta$ steps.

Proof. Since $\sigma' \le 1/4$ and $|T'| \le L_T$, $|F_i(z) - F_i(z')| \le \sum_j |W_{ij}|\, L_T \frac{1}{4}|z_j - z'_j|$, hence $\|F(z) - F(z')\|_\infty \le q\|z - z'\|_\infty$ on the free coordinates (the clamped coordinate does not move). The damped map is the convex combination $(1-\eta)\mathrm{id} + \eta F$, with constant $1 - \eta + \eta q = q_\eta$. Banach's fixed-point theorem gives existence, uniqueness and the first bound; the second follows from the triangle inequality $\|z^t - z^*\| \le \|z^t - z^{t+1}\| + q_\eta\|z^t - z^*\|$ and the bound on $\|z^{t+1} - z^t\| \le q_\eta\|z^t - z^{t-1}\|$.

Corollary 1 (Certified parametrisation). *GraphNet2 sets $\|W\|_\infty = \gamma\,\sigma(\varrho)\,4/L_T(k)$ with $\gamma \le \frac{1}{2}$ (incoming weights normalised to sum one, then scaled by a learnable gain through $\sigma(\varrho)$). Since $\|W_{RR}\|_\infty \le \|W\|_\infty$, $q \le \gamma \le \frac{1}{2}$ for every value of $k \in [0.05, 1]$ and of the learnable parameters, so Theorem 1 holds for every model the optimiser can reach. With $\eta = \frac{1}{2}$, $q_\eta \le \frac{3}{4}$.*

Theorem 2 (Multi-hop response is an absolutely convergent walk sum). *Assume $\rho(|W|) < 1$ (enforced by the scaling in the construction of W). Then for every clamped gene g and free set R: (i) $\rho(|W_{RR}|) \le \rho(|W|) < 1$, so $I - W_{RR}$ is invertible; (ii) $\sum_{k\ge 0} W_{RR}^k W_{Rg}\Delta x_g$ converges absolutely to $(I - W_{RR})^{-1}W_{Rg}\Delta x_g$, so the hop contributions $\Delta x^{(k)} = W_{RR}^{k-1}W_{Rg}\Delta x_g$ and the contributions of individual walks are well defined and may be summed in any order; (iii) with $v = (I - |W|)^{-1}\mathbf{1} \ge \mathbf{1}$ and $q_v := 1 - 1/\max_i v_i < 1$, truncating after K hops leaves an error $\|\sum_{k>K}\Delta x^{(k)}\|_v \le \frac{q_v^K}{1-q_v}\|W_{Rg}\Delta x_g\|_v$, where $\|x\|_v = \max_i |x_i|/v_i$.*

Proof. (i) $|W_{RR}|$ is a principal block of $|W|$; for non-negative matrices the spectral radius is monotone under entry-wise domination and under taking principal blocks (Perron–Frobenius), and $\rho(W_{RR}) \le \rho(|W_{RR}|)$. (ii) $|W_{RR}^k b| \le |W_{RR}|^k|b|$ and $\sum_k |W_{RR}|^k$ converges when $\rho < 1$; absolute convergence permits rearrangement (Fubini) into hops and into walks. (iii) $(I - |W|)^{-1} = \sum_k |W|^k \ge I$ entry-wise, so $v \ge \mathbf{1}$ and $|W|v = v - \mathbf{1} \le (1 - 1/\max v)\,v$. For any x, $|W_{RR}x| \le |W_{RR}||x| \le \|x\|_v|W|v \le q_v\|x\|_v v$ on R, so $\|W_{RR}x\|_v \le q_v\|x\|_v$ and the tail of the geometric series follows.

Remark 1 (Why $\rho(|W|)$). The signed spectral radius does not suffice: $W = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ has $\rho(W) = 0$, but the block $W_{RR} = (1)$ for $R = \{1\}$ has $\rho = 1$ and $I - W_{RR}$ is singular. Nor is $\|W\|_\infty < 1$ needed: on the curated graphs the total incoming weight is $\|W\|_\infty = 5.3$ (K562) and 8.0 (RPE1) after scaling to $\rho(|W|) = 0.90$, while $q_v = 0.977$ and 0.983.

Proposition 2 (SCC-aware execution). *Under Theorem 1, iterating the strongly connected components of the free graph in topological order (a single update for acyclic singletons, a local damped iteration for feedback components with already settled inputs) converges to the same unique fixed point as the global iteration; genes not reachable from g stay at their baseline.*

Proof. Each block map is a restriction of F with the constant of a principal block ($\leq q_\eta$), hence a contraction for fixed inputs. The block solutions, composed in topological order, satisfy every equation of the full system, and the fixed point is unique, so the two coincide. A gene without a path from g has all parents either unreachable or at baseline, so x_i^0 solves its equation, and by uniqueness equals the fixed point.

Proposition 3 (Response-propagation kernels). *For the association graph let $W = AD^{-1}$ (column-stochastic) and $S_\alpha = \alpha(I-(1-\alpha)W)^{-1}$, $\alpha \in (0, 1]$. The series $\sum_k((1-\alpha)W)^k$ converges geometrically with ratio $1-\alpha$, every column of S_α is a probability vector (non-negative, sum one), and the symmetric normalisation $D^{-1/2}AD^{-1/2}$ has spectrum in $[-1, 1]$, so $\rho((1-\alpha)W_{\text{sym}}) \leq 1-\alpha$. The heat kernel e^{-tL}, $L = D - A$, exists for all $t \geq 0$. The propagated prediction $\sum_j S_{pj}\Delta_j / \sum_j S_{pj}$ is a convex combination of training responses and therefore stays in their convex hull.*

Proof. $\|(1-\alpha)W\|_1 = 1-\alpha < 1$ because the columns of W sum to one; the column sums of S_α are $\alpha\sum_k(1-\alpha)^k = 1$; non-negativity follows from the series. L is positive semi-definite, so e^{-tL} is a contraction semigroup.

Remark 2 (What is not claimed). Convergence is not correctness. Endpoint data do not identify dynamics, the fixed point is a steady-state description, and the multi-hop signs beyond one hop are at chance on held-out targets (Section 5.4); the theorems guarantee that the quantity we explain is well defined, not that it matches biology.

The restart-walk kernels have column sums equal to one to within 2×10^{-14} for $\alpha \in \{0.1, 0.5, 0.9\}$.

Table 6. Convergence certificates on the curated signed graphs (all values generated from the run). Linear operator: $\rho(|W|)$ before scaling, scaled value, and Neumann terms needed for 10^{-6} (median/max over 40 clamped genes); q_v and the largest ratio of observed truncation error to the bound of Theorem 2(iii) (≤ 1 confirms the bound). Certified Hill-form operator at extreme and default parameters: contraction constants, steps to 10^{-6} from four random starts, spread of the solutions and fixed-point residual.

	K562	RPE1
$\rho(\|W\|)$ raw $\rightarrow$ scaled	8.5 $\rightarrow$ 0.90	9.3 $\rightarrow$ 0.90
Neumann terms to 10^{-6} (median / max)	12 / 20	11 / 17
q_v; truncation-bound ratio	0.977; 0.019	0.983; 0.014
q (max over parameters); q_η	0.500; 0.750	0.500; 0.750
Steps to 10^{-6} (max)	36	44
Spread of solutions; residual	2.2×10^{-6}; 6.6×10^{-7}	3.0×10^{-6}; 5.4×10^{-7}